

MADHUSUDAN G

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SUMMARY

AI & Data Science undergraduate at MSRIT with experience in machine learning, AI research, computer vision, and developing real-world AI systems involving Edge AI and NLP.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript

AI/ML: TensorFlow, Keras, Scikit-learn, OpenCV, SpaCy, Pandas, NumPy

Web & Tools: Flask, HTML/CSS, Git, Power BI, Postman

Core CS: Data Structures, Graph Algorithms, Complexity Analysis

RESEARCH

Women Safety Heatmap & Secure Routing Framework using GeoAI

Co-Author — NCRIE Conference — Publication Pending

- Developed a framework integrating urban datasets to identify high-risk zones using predictive safety heatmaps and GeoAI-based route planning.
- Presented findings at NCRIE Conference.

WORK EXPERIENCE

AI & ML Intern

Sep 2025 – Nov 2025

Elevate Labs (Remote)

- Built ML pipelines using TensorFlow, Keras, and Scikit-learn for preprocessing, training, and evaluation.
- Improved model performance through optimization and benchmarking.

PROJECTS

BlindSight: Edge-AI Navigation Assistant:

YOLOv8, Raspberry Pi 5, OpenCV

- Developed an offline wearable navigation assistant for visually impaired users using real-time obstacle detection.
- Integrated YOLO, depth estimation, and voice guidance for hazard awareness.

Music Genre Classification (CNN):

Python, Keras, Librosa

- Developed a CNN trained on MFCC audio features for music genre prediction.

Kannada OCR Genesis:

Tesseract OCR, OpenCV, Gemini API

- Built an AI-based Kannada old-text digitization system with OCR and Gemini-based correction.

EDUCATION

Ramaiah Institute of Technology (MSRIT)

Bachelor of Engineering in Artificial Intelligence & Data Science

CGPA: 6.07/10

Bengaluru, India

Expected June 2027

Seshadripuram Public School

ISC, PCMB; Percentage: 55%

Karnataka, India

March 2023

Akash International School

ICSE, Science; Percentage: 68.1%

Karnataka, India

March 2021

ACHIEVEMENTS & LEADERSHIP

- Aspire Leaders Program Graduate selected from 36,000+ global applicants.
- Research & Documentation Lead, IEEE CIS RIT-B; managing documentation and contributing to Women Safety Heatmap research.
- Management Team Member, Team Volante.
- Participated in Inceptrix 2025 Hackathon and Hack-A-War'26(24hr Hackathon) by E-Cell, RIT.

CERTIFICATIONS

Applied Generative AI (Infosys) | Data Science Foundations (IBM) | AWS Solutions Architecture (Forage) | Postman API Fundamentals | SnowPro Associate: Platform (Snowflake)